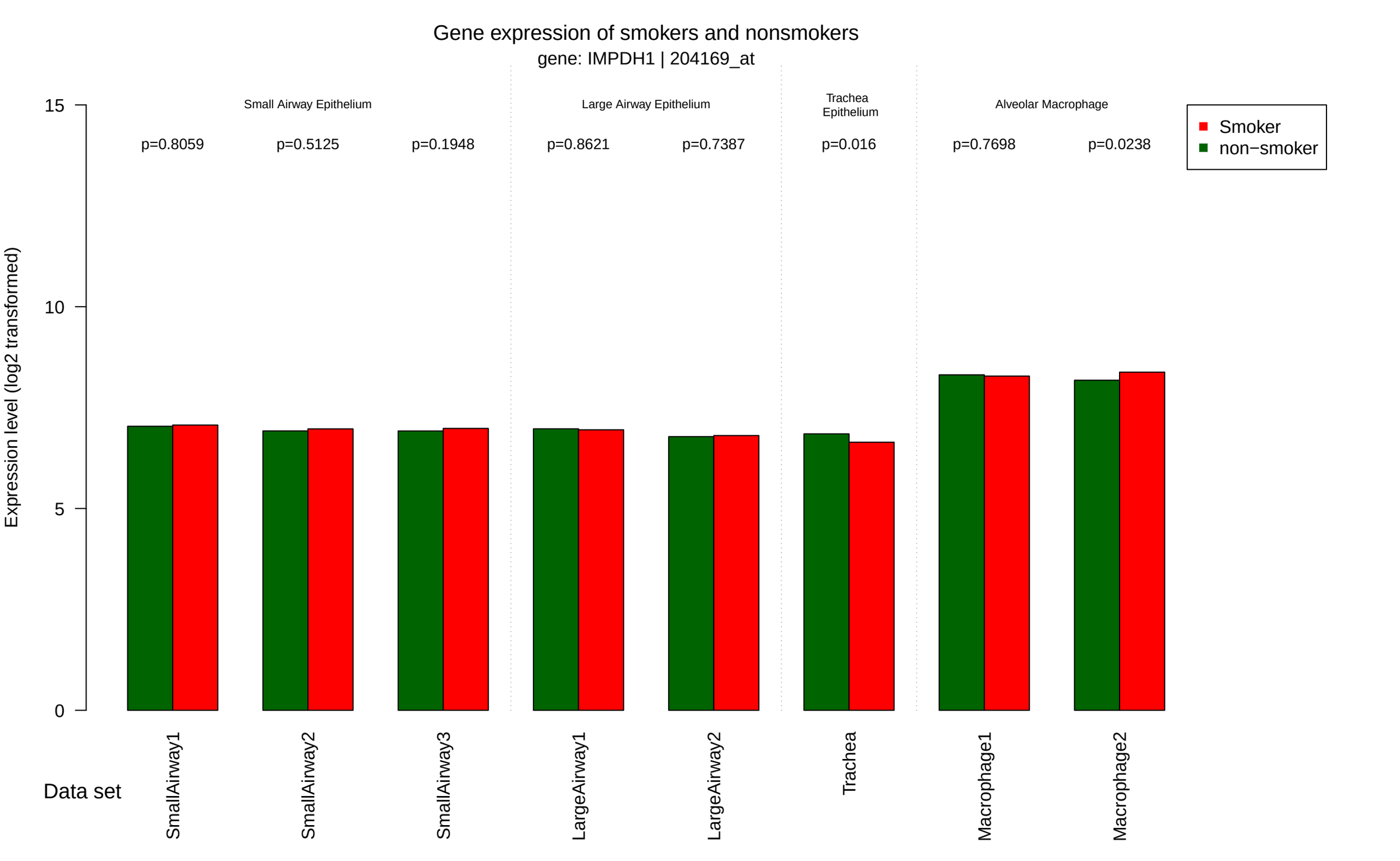
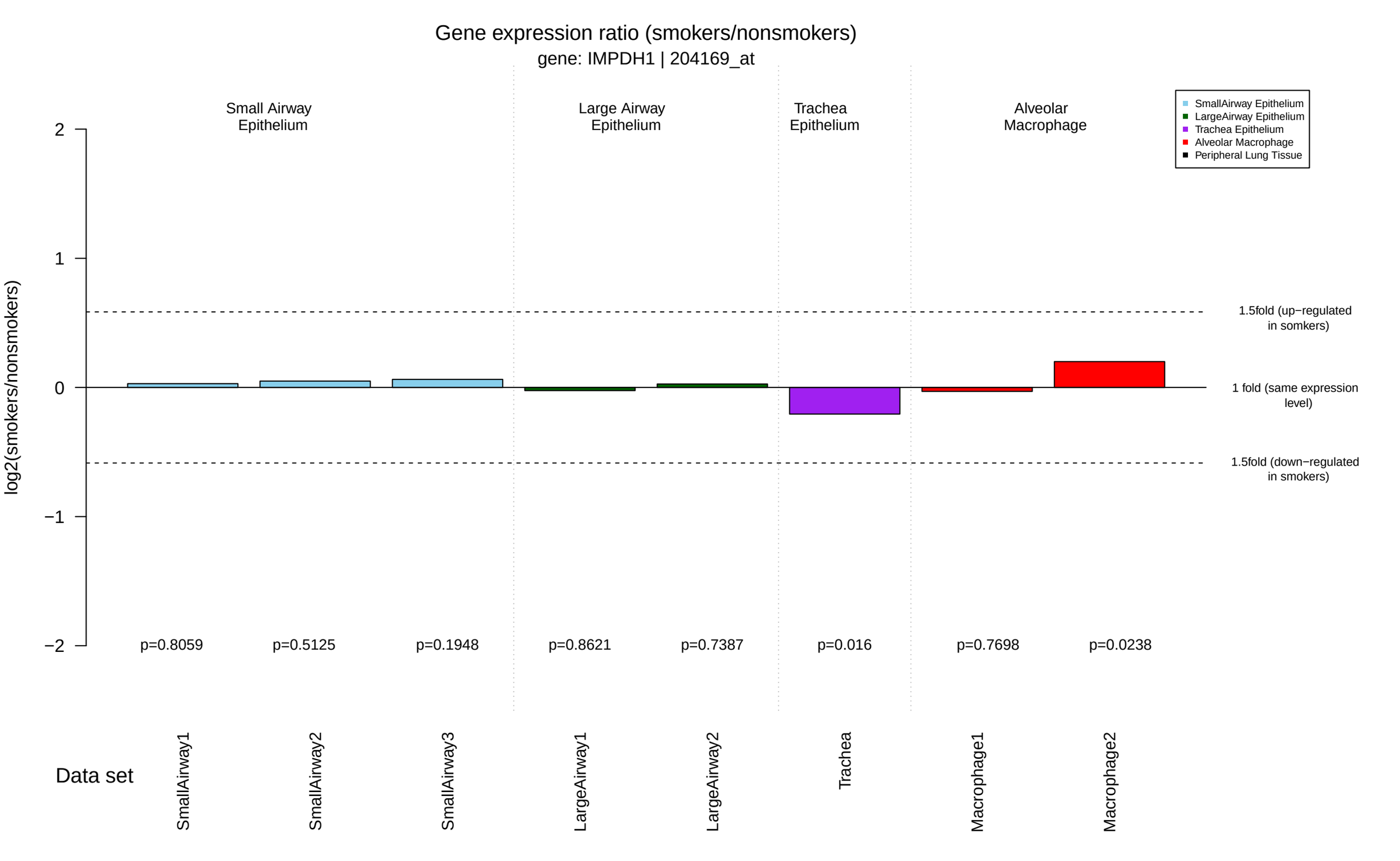
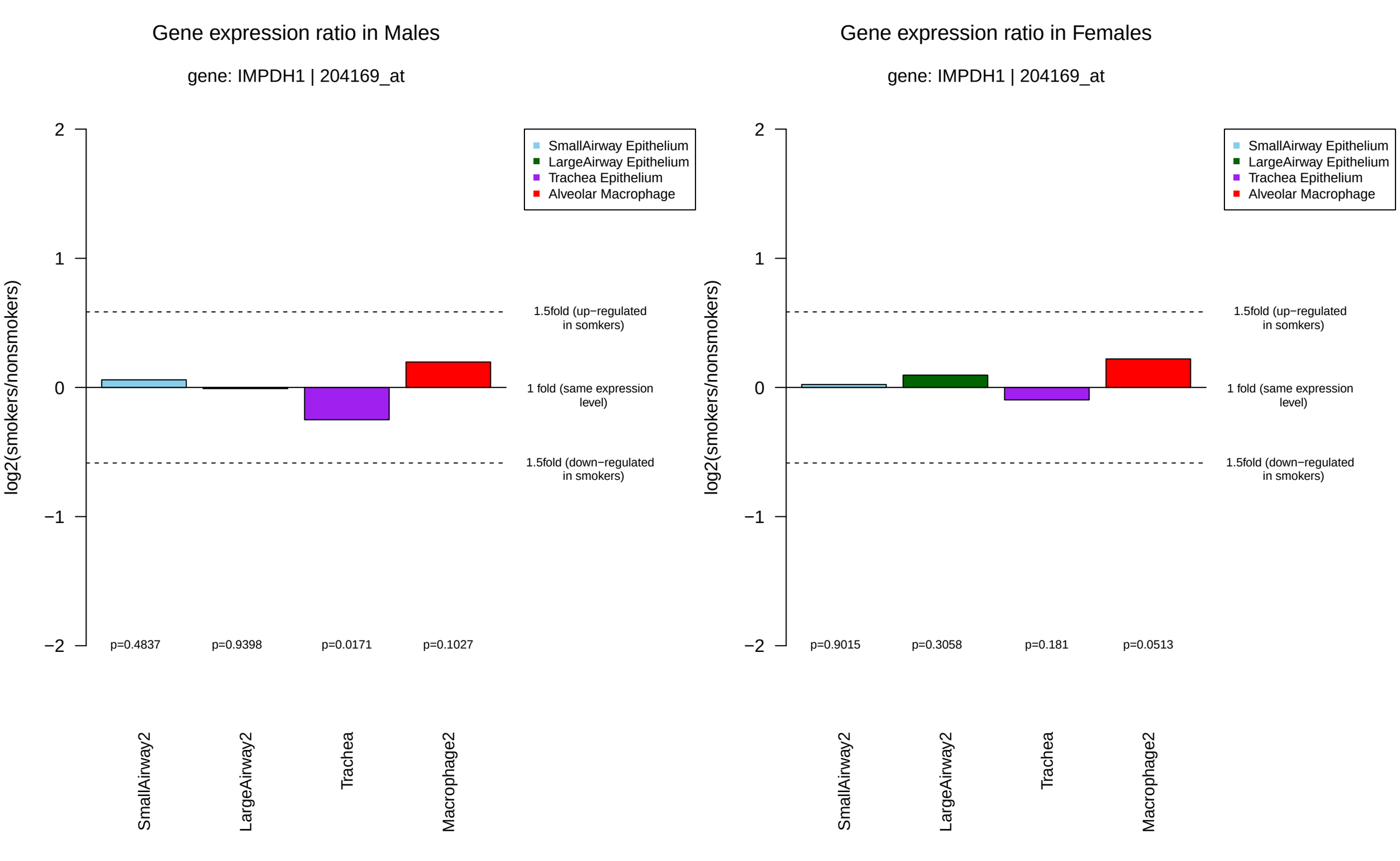
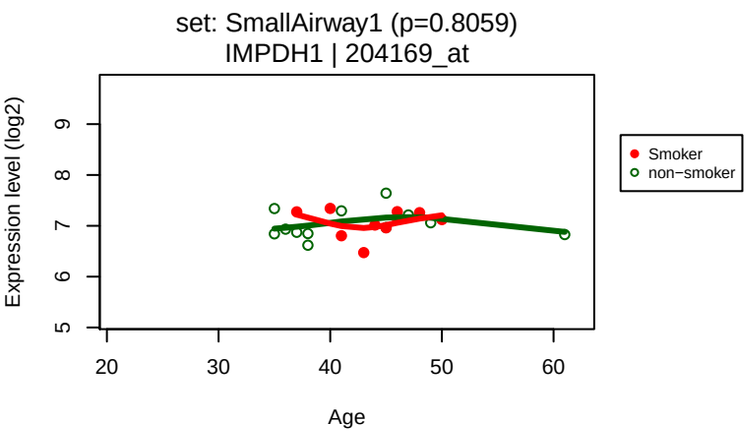
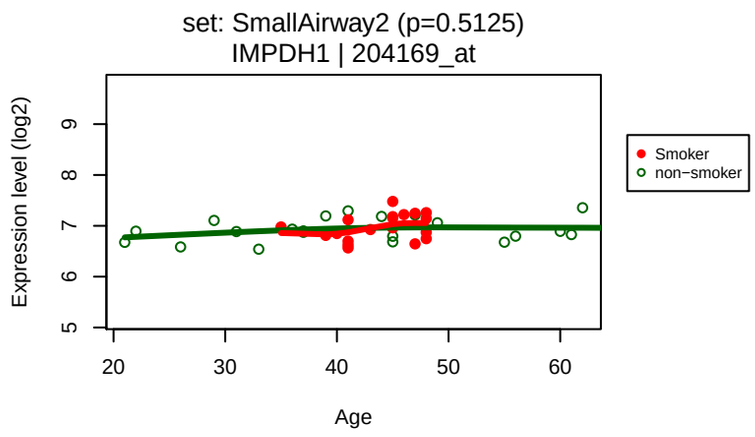
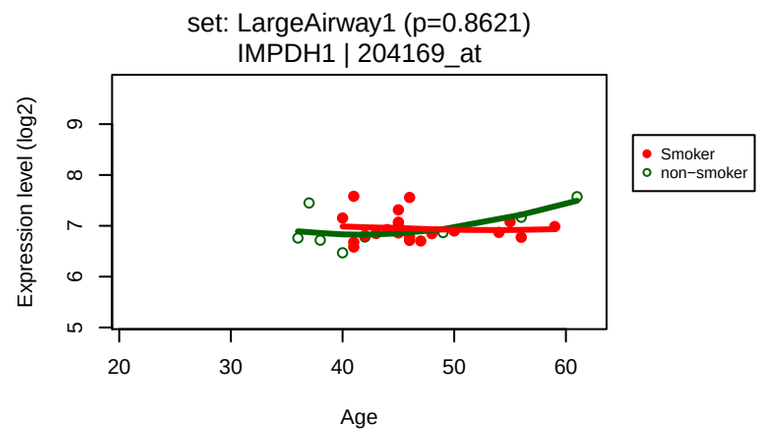

Gene expression of smokers and nonsmokers
gene: IMPDH1 | 204169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8059
p=0.5125
p=0.1948
p=0.8621
p=0.7387
p=0.016
p=0.7698
p=0.0238
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

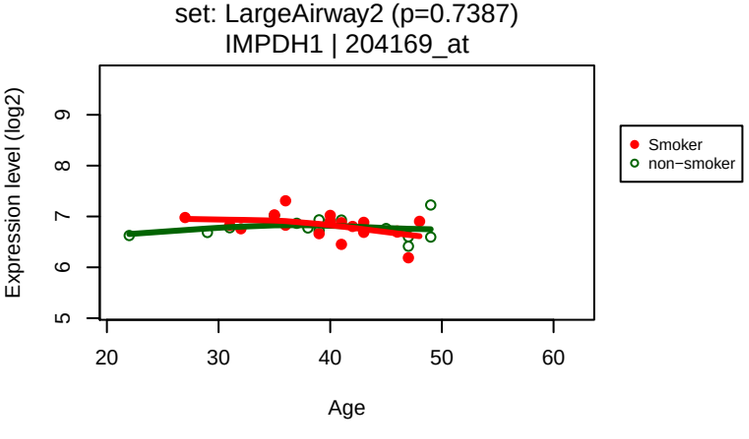

Gene expression ratio (smokers/nonsmokers)
gene: IMPDH1 | 204169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8059
p=0.5125
p=0.1948
p=0.8621
p=0.7387
p=0.016
p=0.7698
p=0.0238
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

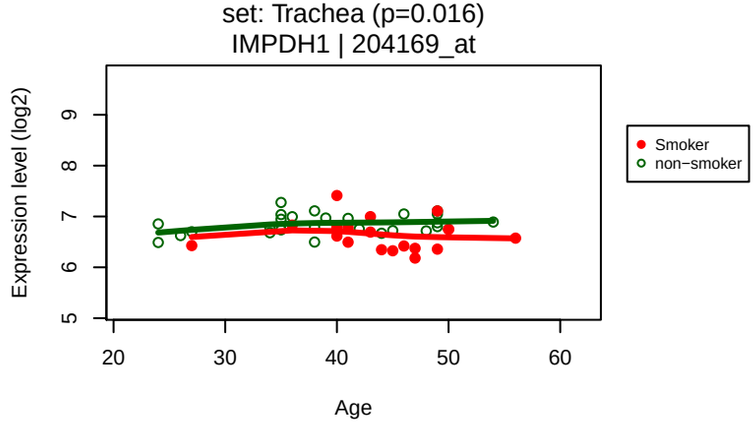

Gene expression ratio in Males
gene: IMPDH1 | 204169_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4837
p=0.9398
p=0.0171
p=0.1027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPDH1 | 204169_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9015
p=0.3058
p=0.181
p=0.0513
SmallAirway2
LargeAirway2
Trachea
Macrophage2

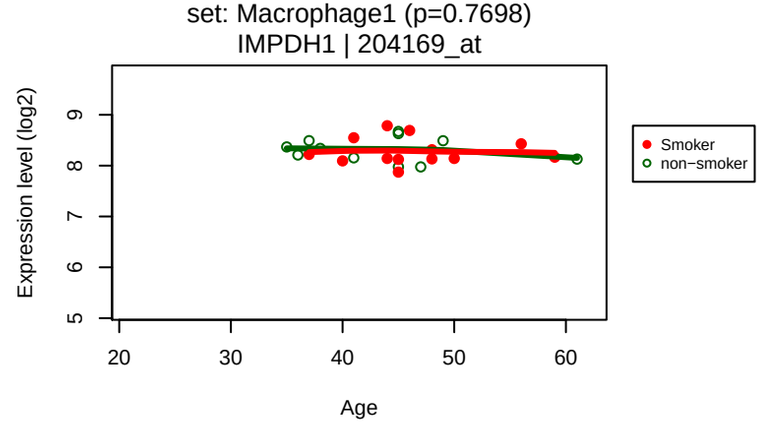

set: SmallAirway1 (p=0.8059)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker

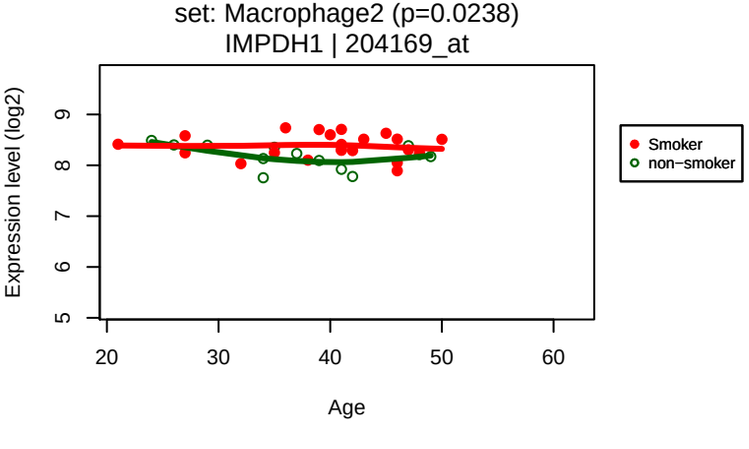

set: SmallAirway2 (p=0.5125)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker

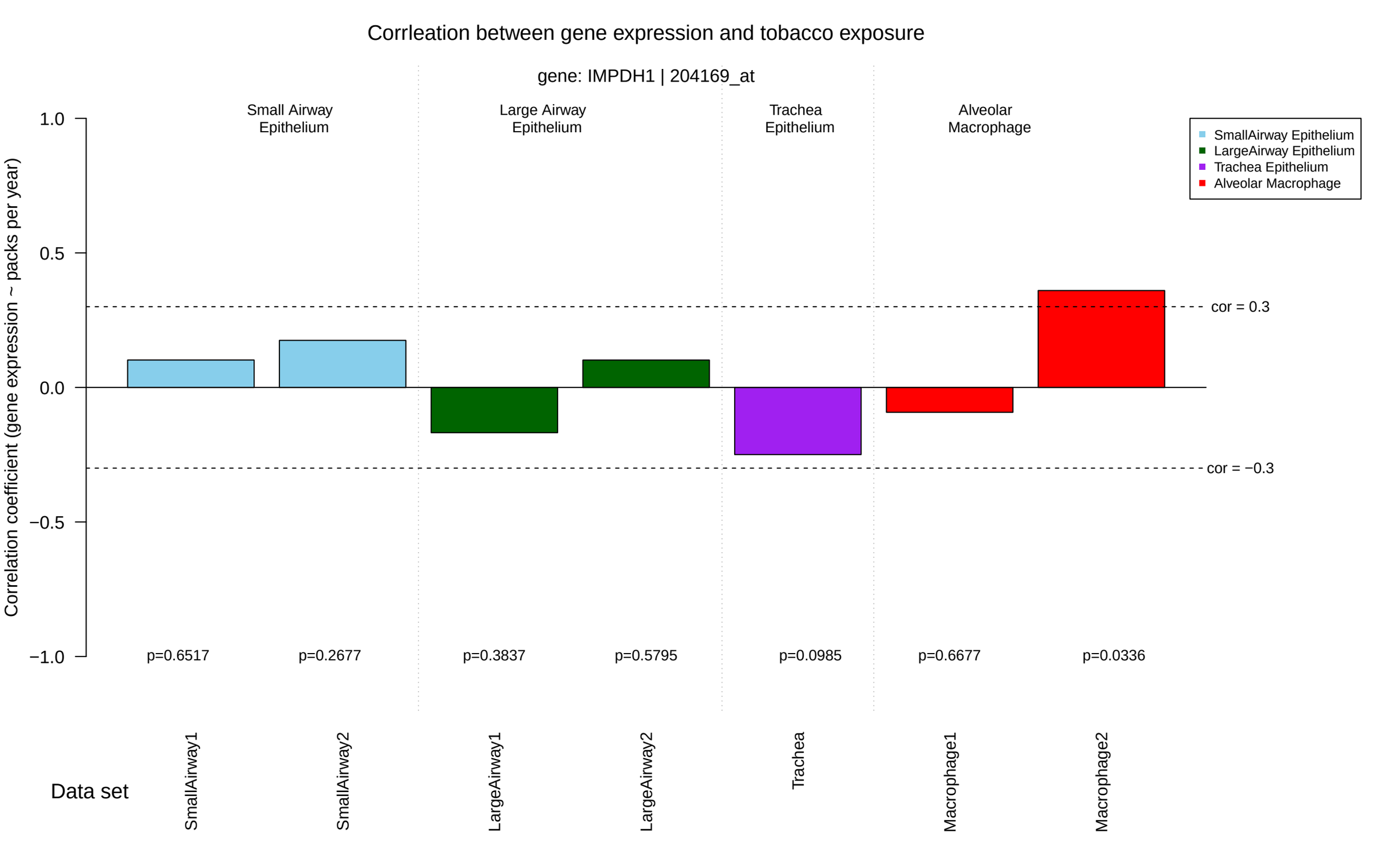

set: LargeAirway1 (p=0.8621)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7387)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.016)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7698)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0238)
IMPDH1 | 204169_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPDH1 | 204169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6517
p=0.2677
p=0.3837
p=0.5795
p=0.0985
p=0.6677
p=0.0336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2